INCH-POUND

MIL-S-10987E(ME)
2 December 1991
SUPERSEDING
MIL-S-10987D(ME)
9 May 1977

MILITARY SPECIFICATION

SCALE, PLOTTING, COORDINATE

(PLASTIC, SQUARE, 1:25,000 SCALE AND 1:50,000 SCALE, GRADUATED IN METERS)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a plastic coordinate scale for use in plotting position of guns, targets, and other reference points on artillery-fire control sheets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6675

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SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13217E7735	- Scale, Coordinate.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JDT, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation. (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Straping, Flat Steel and Seals.
D 4675 - Selection and use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The scale shall be in accordance with TA13217E7735 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operating of the scale. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The scale, plotting, coordinate fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which this item may be exposed.

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3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the scale, plotting, coordinate may be newly fabricated from recovered materials to the maximum extent practicable, provided the scale produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the scale.

3.4 Graduation lines. The graduation lines, numerals, letters, and filling media shall be as specified in accordance with the applicable drawing, and shall be tested for accuracy as specified in 4.5.2.

3.5 Identification marking. The scale shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.6 Workmanship. The scale shall have no rough edges, burrs, cracks, chips, or scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article scale when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table II-A. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected as specified in 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The scale shall be examined as specified herein for the following defects:

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Major

101. Not in conformance to dimensions as shown on the drawings (see 3.1.1).
102. Finish not as specified on drawings (see 3.1.1).
103. Materials not as specified (see 3.3).
104. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.3.1).
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
106. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the scale (see 3.3.4).
108. Graduation lines, numerals, letters, and filling media not in accordance with the applicable drawings (see 3.4).
109. Identification marking missing, incomplete or illegible (see 3.5).
110. Workmanship not as specified (see 3.6).

4.5.2 Test. The accurate spacing of the graduation lines shall be checked at 77 ^{0}F $\pm$5 ^{0}F, by means of a beam compass or dividers and graduated test bar with accuracy traceable to the National Bureau of Standards. The scale graduations should be placed against the approved test bar and compared optically with a direct-measuring microscope or a direct-reading optical comparator accurate to 0.001 inch. Nonconformance to 3.4 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Sample size shall be determined by using MIL-STD-105, table I and table II-A. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

111. Preservation not as specified for level A, B, or C (see 5.1).
112. Consolidation not as specified for level A or B (see 5.1.1.1).
113. Packing not as specified for level A, B or C (see 5.2).
114. Gross weight exceeds 200 pounds for level A (see 5.2.1).

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115. Strapping not as specified for level A (see 5.2.1).
116. Marking missing, illegible, incorrect, or incomplete for levels A, B or C (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or level C, as specified (see 6.2).

5.1.1 Level A. Each scale shall be preserved in accordance with MIL-P-116, method IC-2, using the contractor's commercial box or container, or a close-fitting box conforming to PPP-B-566, variety 1, style, type, and class optional.

5.1.1.1 Consolidation. Scales, preserved as specified in 5.1.1, shall be consolidated in close-fitting boxes conforming to PPP-B-636, W6c, in quantities not to exceed the weight limitation of the box specification.

5.1.2 Level B. Each scale shall be preserved and consolidated as specified in 5.1.1 and 5.1.1.1, except boxes for consolidation shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125 or 175.

5.1.3 Level C. Each scale shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, level B, or level C, as specified (see 6.2).

5.2.1 Level A. Scales, preserved as specified in 5.1, shall be packed together in quantities not to exceed 200 pounds in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Scales, preserved as specified in 5.1, shall be packed together in close-fitting fiberboard boxes conforming to PPP-B-636, V3c, style optional. The gross weight or size of each box shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Scales, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking for shipment and storage shall be in accordance with MIL-STD-1190.

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6. NOTES

(This section contains information of general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The scale is intended for use in reading and plotting coordinates on 1:25,000 and 1:50,000 scale gridded maps and photographs.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
c. When a first article is required for inspection and approval, and the number of units required (see 3.2).
d. Level of preservation and level of packing required (see 5.1 and 5.2).
e. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject (key word) listing.

Scale graduations
Test bar
Direct measuring microscope
Optical comparator

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
DLA - GS

Preparing Activity:
Army - ME

Project 6675-A939

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-10987E(ME)	91/12/02

3. DOCUMENT TITLE
Scale, Plotting, Coordinate (Plastic, Square, 1:25000 Scale & 1:50,000 Scale, Graduated Meter

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
	(1) Commercial (2) AUTOVON (If applicable)	c. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717

c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
US Army Belvoir RDE Center ATTN: STRBE-TSE Ft Belvoir VA 22060-5606	Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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